Markups and Public Procurement: Evidence from Czech Construction Tenders

Marek Chadim[1]

Writing Sample for Ph.D. Applications

Abridged and revised from the MSc thesis, short version

November 2026

Abstract

This paper analyzes the effect of public procurement on firm markups, using a panel dataset of Czech construction firms from 2006 to 2021. Markups are estimated through a structural framework, while biases are addressed using a selection on observables design and models for unobserved factors. Propensity score-based estimations indicate that firm markups increase by approximately 15% during contract years. A temporal analysis, employing synthetic control and matrix completion methods, reveals that treatment effects decline from around 30% in 2006 to 10% in 2021. These patterns are consistent with institutional improvements in the Czech Republic and offer empirical evidence of increasing efficiency in public spending.

Keywords: Models with Panel Data; Firm Behavior: Empirical Analysis; Procurement; Construction

JEL: C23, D22, H57, L74

[1]Predoctoral Fellow, Tobin Center for Economic Policy, Yale University (marek.chadim@yale.edu; https://marek-chadim.github.io). This paper is abridged and revised from my MSc thesis, submitted to the Department of Economics, Stockholm School of Economics (defended December 2024, awarded the highest grade) and awarded First Place in the 32nd Young Economist of the Year competition of the Czech Economic Society, 2025. The thesis as examined is in the school's library: https://hhs.primo.exlibrisgroup.com/permalink/46SSOE_INST/hejhp1/alma991001619398206056; Methods A to C and the appendix tables cited in the text are there. Section 3.1 adapts the export-status analysis of De Loecker and Warzynski (2012), and Section 3.2 applies the workflow of Imbens and Xu (2024) to these data; its design, estimator set and figures follow theirs. Of Section 3.3 only the last result is reproduced. I am grateful to my thesis supervisor, Jaakko Meriläinen, for invaluable guidance; to Matěj Bajgar, Mitch Downey, and David Schönholzer for their helpful feedback; and to Jiří Skuhrovec of Datlab s.r.o. for providing administrative data on government procurement. The thesis builds on my earlier work at the Institute of Economic Studies, Charles University Prague (https://dspace.cuni.cz/handle/20.500.11956/184831), where I accessed firm financial statement data via MagnusWeb. Replication files: https://github.com/marek-chadim/thesis. This short version keeps the introduction, the data, the cross-section premium, Section 3.2, the last result of Section 3.3 and the conclusion.

1 Introduction

Public procurement constitutes approximately 12% of GDP in OECD countries, representing a substantial share of government expenditure (OECD, 2021). Ensuring its efficiency is vital for reducing resource waste and maximizing taxpayer value. Existing literature has examined key issues in public procurement, including discretion, political favoritism, and limited competition. However, much of this research has focused on comparing tenders, with relatively little attention devoted to differences between public procurement and private market outcomes.

This paper fills this gap by constructing a novel dataset and proposing methodological solutions to the identification challenges inherent in studying the dynamics of firm markups as firms enter and exit public procurement markets. To the best of my knowledge, this is the first study to provide credible econometric evidence on the relationship between markups and participation in public procurement. The findings contribute to ongoing debates on public procurement efficiency, market competition, and the implications of government contracts for firm-level market power.

Using the methodology of De Loecker and Warzynski (2012), I estimate markups under the assumption that cost-minimizing producers optimize variable input expenditures. This approach provides firm-level markup estimates without relying on assumptions about firms' competitive behavior in product markets. The results document the evolution of aggregate markups, showing a decline from 44% above marginal cost in 2006 to 31% in 2021, driven primarily by firms at the upper end of the markup distribution.

Next, I estimate the public procurement premium using non-experimental methods under the assumption of unconfoundedness, conditioning on information available prior to firms' engagement in public procurement. To mitigate biases arising from observational characteristics, I apply methodological advancements from Imbens and Xu (2024). Placebo tests, which compare the average markups of treated and control firms before the treated firms receive contracts, and sensitivity analyses benchmarking the omitted variable bias further reinforce the robustness of the findings.

Finally, I employ panel data models for binary interventions in longitudinal settings to estimate the causal effect of public procurement on the markups of government contractors, drawing on Arkhangelsky and Imbens (2024). These methods extend traditional difference-in-differences and two-way fixed-effects models by incorporating factor models, interactive fixed-effects models, and synthetic control approaches. The analysis emphasizes heterogeneity in causal effects and uses valid inference techniques suited to cases with a small number of treated units.

Drawing causal inferences from observational data, even with advanced methods, requires strong assumptions. The lack of natural experiments, instrumental variables, or regression discontinuity designs limits the availability of exogenous variation in this study. To address these challenges, I use a selection-on-observables approach, framed within the potential outcomes framework, to model counterfactual markups for government contractors in the absence of public procurement. The stable unit treatment value assumption (SUTVA) excludes interference between units and the dynamic effects of past treatments and outcomes. However, in competitive public procurement markets, violations of SUTVA can complicate the interpretation of causal effects. Despite this, the study prioritizes internal validity through a design-based balanced panel approach while enhancing generalizability by leveraging the full dataset. Balanced panel estimates, based on selection on observables, indicate that procurement raises contracting firms' markups by an average of 15%, with the full panel data model showing a decline in this effect from 30% in 2006 to 10% in 2021.

Industrial organization models with heterogeneous producers and firm specific markups provide potential explanations for these findings. More productive firms may absorb the costs associated with

entering public procurement markets, which could explain why government contractors typically exhibit higher markups. Additionally, contractors may charge higher markups if they supply higher-quality goods produced using higher-quality inputs. Another plausible explanation is that procurement contracts often provide firms with long-term clients, leading to less elastic realized demand and enabling higher markups.

The literature on public procurement also highlights other mechanisms, such as discretion and political favoritism. My findings suggest that the public procurement markup premium has declined over time, consistent with reforms targeting single-bidding practices (Titl, 2023) and increased monitoring associated with European Union co-financing (Baránek and Titl, 2024).

2 Data

I utilize a dataset obtained through licensed[2] access to the MagnusWeb database[3], which provides full company accounts for an unbalanced panel of 1,297 firms active in the Czech construction sector. The dataset contains 7,261 observations covering the period from 2006 to 2021. The sample size accounts for the markup estimation requirement of having at least one lag of data. For robustness of the production function estimates, the top and bottom percentiles of firms are trimmed based on the sales-to-cost-of-goods-sold ratio.

In the Czech Republic, public procurement contracts are awarded under nationwide regulations requiring the publication of contract details by procurers. The dataset includes contracts awarded by central, regional, and municipal governments, as well as government-owned enterprises. A private company[4] corrected and cleaned the raw data to ensure its accuracy. I link firm financial statements with public procurement records to construct a binary indicator of government tender sales. This classification identifies firms that operate exclusively in the private sector, those that enter public procurement, those that exit it, and those that remain as government contractors.

3 Results

Section 3.1 documents the main patterns of markups in the Czech construction sector and the relationship between markups and public procurement. I validate these results using two treatment effect identification strategies.

Section 3.2 employs a design-based approach relying on unconfoundedness, or selection based on observable financial statement information and pre-procurement markups. In contrast, Section 3.3 focuses on imputing the counterfactual markup evolution for government contracting firms, addressing selection on latent unobservable factors and incorporating synthetic control methods.

3.1 Markups and Public Procurement in the Czech Construction Sector

I measure markups with the production approach of De Loecker and Warzynski (2012). A cost-minimizing firm that chooses a variable input free of adjustment costs sets its markup equal to the output elasticity of that input divided by the input's share of revenue, $\mu_{it} = \theta^v_{it}/\alpha^v_{it}$. The share is read from the accounts. The elasticity comes from a translog production function estimated on deflated

[2]Charles University Prague, Faculty of Social Sciences, Institute of Economic Studies

[3]Dun & Bradstreet Czech Republic, a.s.

[4]Datlab s.r.o.

sales with the control function approach of Ackerberg et al. (2015), in which a firm's public procurement status enters the input demand and the law of motion for productivity. Details are in Methods A of the thesis.

Do Government Contractors Have Different Markups? I examine whether government contractors systematically have different markups compared to private-sector firms by using firm-specific markups in a regression framework. Specifically, I estimate the percentage difference in markups between government contractors and private-sector firms.

This approach ensures robustness even if the variable inputs used to compute markups are subject to adjustment costs, provided that such costs do not disproportionately affect government contractors.

The regression specification is as follows:

$$\ln \mu_{it} = \delta_0 + \delta_1 \, pp_{it} + \mathbf{b}'_{it}\sigma + \nu_{it}, \tag{1}$$

where pp_{it} is a dummy variable indicating a firm's public procurement status, and δ_1 measures the percentage markup premium for government contractors. The vector $\mathbf{b}_{it}$ includes control variables[5], with σ as their corresponding coefficients.

It is important to note that δ_1 is not interpreted as a causal parameter. Instead, this specification identifies whether government contractors, on average, have different markups. To the best of my knowledge, this relationship has not previously been documented, making these findings a novel contribution. While the coefficients of the control variables are not the primary focus of this analysis, I revisit the correlation between markups and other economic factors later in the paper. The regression is estimated on the full sample of construction firms with full year–subindustry interaction dummies. The percentage premium parameter estimate, $\hat{\delta}_1$, is 0.149, with a standard error of 0.003.

These results suggest that government contractors, on average, charge 14.9% higher markups than private-sector firms.

3.2 Estimation of the Public Procurement Effect Under Unconfoundedness

This section estimates the effect of holding a public contract on a firm's markup in the contract year, under the assumption that selection into contracts depends only on what I observe. The design follows the workflow that Imbens and Xu (2024) recommend for observational studies: estimate the propensity score flexibly, check overlap and trim, compare several estimators including doubly robust ones, and test the identifying assumption with placebo outcomes and a sensitivity analysis. The two assumptions behind it, unconfoundedness and overlap, are stated in Methods B.

I do not observe how contracts are assigned. What I do observe is each firm's markup history, which I use both to adjust for selection and to test the adjustment. To have a complete history for every firm, I restrict the sample to a balanced panel as a design-based decision: firms observed in all 16 years, excluding those that already held public contracts in 2006. This leaves 26 firms. The covariates need four lags, so the estimation sample covers 2010–2021 and has 312 firm-year observations.

The treatment is binary: $W_{it} = 1$ if firm i has revenue from public tenders in year t. There are 151 treated firm-years. The 161 control firm-years are those without such revenue: 60 belong to five firms that hold no public contract in 2010–2021, and 101 are years in which the other 21 firms hold none. The 21 firms differ in how often they contract: two do so in a single year, three in all twelve, and the

[5]Control variables include variable input and capital use to account for differences in size and factor intensity, along with 47 year–subindustry interaction dummies to capture aggregate markup trends.

median firm in seven. The contracts in the sample total 25.9 billion Czech crowns over 2010–2021, about 1.2 billion USD at the 2021 exchange rate (1 USD = 21.5 CZK).

The outcome is the log markup in the contract year, $Y_{i,0}$; I also report the two following years, $Y_{i,1}$ and $Y_{i,2}$. The windows for testing and for adjustment are those of Imbens and Xu's lottery application. The log markups of the three years before the contract, $Y_{i,-3}, Y_{i,-2}, Y_{i,-1}$, and their average are placebo outcomes: a contract cannot affect them, so an estimated effect on them would signal selection that the adjustment has missed. The log markups of the three years before that, $Y_{i,-6}, Y_{i,-5}, Y_{i,-4}$, are covariates, together with the firm's sales, cost of goods sold, assets and number of employees four years earlier, its public procurement status in each of the four preceding years, the calendar year and the NACE 2-digit division. Where a lagged or later markup falls outside the panel, the code sets it to zero; this affects the first two years of the covariates and the last two years of the later outcomes.

Table 1 compares treated and control firm-years on covariates measured four years earlier. Contract years belong to larger firms with higher markups: the difference in lagged log markups is 0.64 pooled standard deviations, and the difference in employment is 0.99. Differences above 0.25 are generally considered substantial (Imbens and Wooldridge, 2009), and with gaps of this size a linear regression leans on extrapolation and its result depends on functional form. Each estimator below uses the control firm-years to predict what a treated firm-year's markup would have been without the contract, $\mu_0(x) \equiv E[Y_i(0) \mid X_i = x]$.

Table 1: Pretreatment Summary Statistics

Observations Stratified by Engagement in Public Procurement	$\{i,t;W_{it}=0\}$ N=161 **mean (s. d.)**	$\{i,t;W_{it}=1\}$ N=151 **mean (s. d.)**	**diff / sd**
No. Employees	52.57 (42.03)	104.96 (61.67)	0.993
$\ln(\text{Sales})_{t-4}$	18.52 (1.01)	19.27 (0.85)	0.799
$\ln(\text{Costs})_{t-4}$	17.89 (0.95)	18.52 (0.82)	0.713
$\ln(\text{Markup})_{t-4}$	0.24 (0.13)	0.34 (0.17)	0.640
$\ln(\text{Assets})_{t-4}$	16.05 (1.68)	16.97 (1.37)	0.603
NACE 42	0.04 (0.20)	0.11 (0.32)	0.259
NACE 43	0.40 (0.49)	0.44 (0.50)	0.081

Note: Firm-years from the balanced panel of 26 firms; 13 firms already active in public procurement in 2006 are excluded. The years 2006 to 2009 supply the lags. $N \times T = 312$ observations in 2010–2021. The measure diff/sd is the absolute difference in means divided by the pooled standard deviation:
$\text{diff/sd} = \left|\bar{X}_0 - \bar{X}_1\right| \left(\sqrt{(s_0^2 + s_1^2)/2}\right)^{-1}$.

I estimate propensity scores with a generalized random forest. Most treated observations have scores that control observations also reach, but treated observations with scores above 0.8 have almost no counterparts.

Following their trimming procedure, I drop observations with a score above 0.8, which removes 53 treated and 8 control firm-years, and then match each remaining treated observation to one control observation on the re-estimated score, without replacement. The matched sample has 97 treated and 97 control firm-years.

Table 2 reports the average effect on the treated in the matched sample. Its rows are the eleven estimators that Imbens and Xu (2024) compare: the raw difference in means; three outcome models (linear regression, Oaxaca–Blinder, and a generalized random forest, GRF); two matching estimators

(five nearest neighbors with bias correction, and the propensity score); three weighting estimators (inverse probability weights from GRF scores, covariate balancing propensity scores, and entropy balancing); and two doubly robust estimators (double machine learning with an elastic net, and augmented inverse probability weighting with GRF, AIPW-GRF). Across the eleven rows the contract-year estimate lies between 0.12 and 0.16, about 15% on average, in line with the descriptive premium of Section 3.1: holding a public contract raises a firm's markup in the contract year.

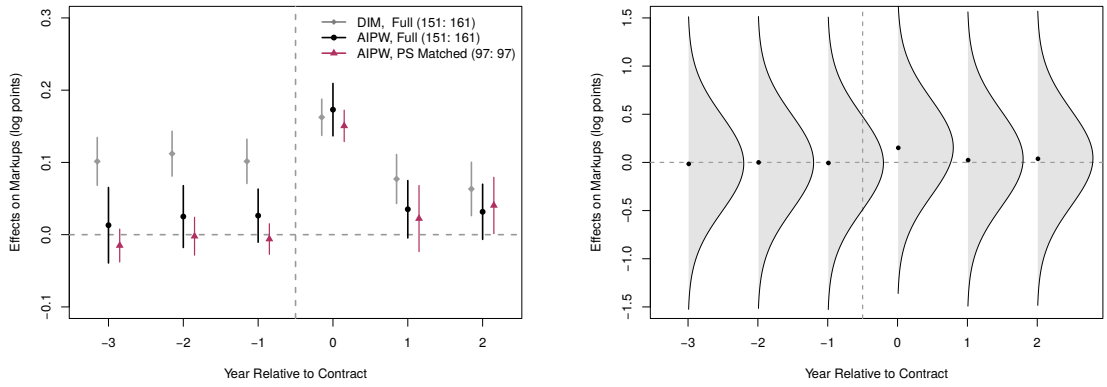
The last two rows are the ones to read for the placebo test: their estimates for the pre-contract markup are -0.01 and 0.00. The standard errors are heteroskedasticity-robust and treat firm-years as independent; with 26 firms, errors clustered by firm would be wider.

Table 2: Contract-Year and Placebo Estimates, Matched Sample

Effect on Markups	**Contract**	**Pre-Contract Average**
Difference-in-Means	0.12 (0.02)	0.03 (0.02)
Regression	0.16 (0.01)	-0.00 (0.01)
Oaxaca Blinder	0.15 (0.01)	0.00 (0.02)
GRF	0.13 (0.01)	0.03 (0.01)
NN Matching	0.15 (0.01)	0.01 (0.01)
PS Matching	0.13 (0.01)	-0.00 (0.01)
IPW	0.14 (0.02)	0.01 (0.02)
CBPS	0.15 (0.02)	0.00 (0.02)
Entropy Balancing	0.15 (0.03)	-0.00 (0.02)
DML-ElasticNet	0.16 (0.01)	-0.01 (0.01)
AIPW-GRF	0.15 (0.01)	0.00 (0.01)

Note: ATT estimates in log points on the trimmed and matched sample (propensity scores above 0.8 excluded). First column: log markup in the contract year. Second column: the placebo outcome, the average log markup of the three years before the contract. Robust standard errors in parentheses. Estimator set after Imbens and Xu (2024).

Figure 1: Estimates by Year Relative to the Contract



(a) ATT (b) CATT

Note: Panel (a): ATT in log points by year relative to the contract, with 95% confidence intervals. Gray: difference in means, full sample. Black: AIPW-GRF, full sample. Red: AIPW-GRF, matched sample. Years –3 to –1 are placebo outcomes. Panel (b): spread of the conditional effects (CATT) from AIPW-GRF in the matched sample, by year; the dots are the ATT. Figure design after Imbens and Xu (2024).

Figure 1 shows the same comparison for each year from three before the contract to two after. Before the contract the raw difference in means is already about 0.10, the selection that Table 1 shows, while the AIPW-GRF estimates are close to zero in all three pre-contract years, in the full sample and in the matched one. The plot follows Imbens and Xu's figure for the lottery data and, as they note of theirs, it is not an event study: each point is a separate cross-sectional comparison under unconfoundedness, and no parallel-trends assumption is involved.

The estimate is also stable when I allow for an unobserved confounder. In the sensitivity analysis of Imbens (2003), as extended by Cinelli and Hazlett (2020), the regression estimate with all covariates is 0.159, and a confounder three times as strongly related to treatment and to the outcome as the markup of the year before the contract would leave it at 0.157.

Taken together, the pre-contract estimates give no sign of remaining selection in this sample. That is as far as the test goes: it speaks to selection on past markups, not to anything that changes in the contract year itself.

3.3 Model-based Estimation of the Public Procurement Effect

This section of the thesis extends the analysis from the balanced panel to the full sample and to selection on unobserved factors. It imputes each contractor's markup without the contract from fixed-effects, interactive fixed-effects and matrix completion models (Liu et al., 2024; Athey et al., 2021). On the balanced panel of Section 3.2, the fixed-effects estimate of the contract-year effect is 0.20. Only the last result of the section is reproduced here.

I present the year-aggregated ATT estimates from the matrix completion method in Table 3. The results provide further evidence of a decreasing trend in the public procurement premium over time for firms in the Czech construction sector. Specifically, my estimates suggest that, compared to firms operating only in the private sector, government contractors had markups approximately 30% higher from 2006 to 2009, over 20% higher during the period 2010–2015, and about 10% higher during 2016–2021, only one-third of their initial level.

Table 3: Matrix Completion Year-Aggregated ATT Estimates

Year	ATT	Standard Err.	No. Treated
2006	0.294	0.016	12
2007	0.282	0.022	7
2008	0.300	0.014	8
2009	0.337	0.010	18
2010	0.273	0.017	9
2011	0.241	0.012	11
2012	0.236	0.013	18
2013	0.262	0.010	18
2014	0.301	0.008	20
2015	0.219	0.008	31
2016	0.092	0.008	34
2017	0.105	0.008	77
2018	0.100	0.006	68
2019	0.101	0.006	74
2020	0.118	0.007	78
2021	0.098	0.008	58

4 Conclusion

Using firm-level data on firms in the Czech construction sector, this study employs a method that accommodates diverse price-setting strategies and nonconstant returns to scale to estimate firm-specific markups. It describes the evolution of market power through the distributional characteristics of markups. Between 2006 and 2021, markups declined from 44% to 31% above marginal cost, a decline primarily driven by firms with the highest initial markups. Over time, the distribution of markups became less skewed, with a thinner upper tail.

The decline in Czech construction aggregate markups is linked to the dynamics of the public procurement markup premium. The relationship is validated using policy evaluation designs to account for both observable and unobservable selection bias.

In the balanced sample, the year-of-contract-treatment effect is estimated to be 15% under selection on observable lagged variables, while the same-sample fixed-effects analysis yields a point estimate of 20%. Using both the unconfoundedness approach and the TWFE model to test robustness to alternative identifying assumptions leads to the bracketing property bounds on the estimated causal effect (Angrist and Pischke, 2009, Section 5.3).

The analysis also integrates Synthetic Difference-in-Differences re-weighting and matching on pre-exposure trends to reduce reliance on parallel-trend assumptions. This approach accounts for additive unit-level shifts while utilizing data-driven synthetic control methods.

Finally, the full sample estimation employs interactive fixed effects and matrix completion methods. The framework is validated using diagnostic tests of out-of-sample predictions for untreated potential outcomes, which are robust to model misspecification and overfitting. Subgroup analysis indicates that the overall effect is primarily driven by government contracts awarded in the early years of the sample. The estimated average public procurement effect on government contractors is 30% in 2006 and decreases to 10% by 2021.

These findings contribute to the public procurement literature by establishing a relationship between markups and public procurement. They also provide a benchmark for assessing the competitiveness of government projects relative to those in the private sector. The results provide empirical support for reforms aimed at eliminating single-bidding practices and increasing transparency in the tendering process that prevents favoritism in public procurement. They offer policy-relevant insights, highlighting the role of institutional improvements in enhancing the efficiency of public spending.

Limitations and Future Directions This study has several limitations related to its methodology and the available data, highlighting opportunities for refinement and further research.

A key limitation is the reliance on deflated sales as a proxy for physical quantities in estimating output elasticities. While practical given the data constraints, this approach may introduce omitted variable bias due to unobserved price variations, potentially distorting output elasticity and markup estimates. Access to firm-specific price data or product-level sales would improve the precision of these estimates by disentangling the effects of costs and pricing strategies. Such data would also enable a more detailed analysis of how public procurement influences firm-level pricing behavior.

References

Ackerberg, D. A., Caves, K., and Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6):2411–2451.

Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Arkhangelsky, D. and Imbens, G. (2024). Causal models for longitudinal and panel data: a survey. *The Econometrics Journal*, 27(3):C1–C61.

Athey, S., Bayati, M., Doudchenko, N., Imbens, G., and Khosravi, K. (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536):1716–1730.

Baránek, B. and Titl, V. (2024). The cost of favoritism in public procurement. *The Journal of Law and Economics*, 67(2):445–477.

Cinelli, C. and Hazlett, C. (2020). Making sense of sensitivity: extending omitted variable bias. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 82(1):39–67.

De Loecker, J. and Warzynski, F. (2012). Markups and firm-level export status. *The American Economic Review*, 102(6):2437–2471.

Imbens, G. and Wooldridge, J. (2009). Recent developments in the econometrics of program evaluation. *Journal of Economic Literature*, 47(1):5–86.

Imbens, G. and Xu, Y. (2024). LaLonde (1986) after nearly four decades: lessons learned. arXiv preprint arXiv:2406.00827v2.

Imbens, G. W. (2003). Sensitivity to exogeneity assumptions in program evaluation. *The American Economic Review, Papers and Proceedings*, 93(2):126–132.

Liu, L., Wang, Y., and Xu, Y. (2024). A practical guide to counterfactual estimators for causal inference with time-series cross-sectional data. *American Journal of Political Science*, 68(1):160–176.

OECD (2021). Public procurement. https://www.oecd.org/en/topics/policy-issues/public-procurement.html. Accessed: October 3, 2024.

Titl, V. (2023). The one and only: single-bidding in public procurement. Working papers, Utrecht School of Economics.